AN OPEN IMAGE OF A GROUP-SCHEME HOMOMORPHISM OVER A FIELD IS CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original lemma 0B7N: sources/stacks-chow/tags.txt, source lines 14111–14111. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $\psi : G' \to G$ be a morphism of group schemes over k. If $\psi(G')$ is open in G, then $\psi(G')$ is closed in G.*

Proof. Let $U = \psi(G') \subset G$. Let $Z = G \setminus \psi(G') = G \setminus U$ with the reduced induced closed subscheme structure. By Lemma 0B7N the image of

$$Z \times_k G' \longrightarrow Z \times_k U \longrightarrow G$$

is open (the first arrow is surjective). On the other hand, since ψ is a homomorphism of group schemes, the image of $Z \times_k G' \to G$ is contained in Z (because translation by $\psi(g')$ preserves U for all points g' of G'; small detail omitted). Hence $Z \subset G$ is an open subset (although not necessarily an open subscheme). Thus $U = \psi(G')$ is closed. □

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.